Catalyst. *by Mrs. Rana*

GRADE 9 · GENERAL · TERM 1 · END-OF-TERM PRACTICE

# Mock Exam

Built on the ministry exam structure: 20 multiple-choice questions followed by 5 written questions. Every lesson of the term is tested.

| | |
|---|---|
| **TIME**<br>120 minutes | **PART A**<br>Questions 1–20 · multiple choice · 3 marks each |
| **PART B**<br>Questions 21–25 · written · 8 marks each | **TOTAL**<br>100 marks |

Name _____

Class _____ Date _____

Score _____ / 100

## INSTRUCTIONS

- Answer every question. For Part A choose one answer.
- For Part B show all your steps. Correct steps earn marks.
- Calculators are allowed unless your teacher says otherwise.

LESS EFFORT. FULL MARK.

# Part A

**Questions 1–20 · Choose the correct answer · 3 marks each**

**1** Write a numerical expression for: *two plus four divided by eight.*

A $8 \div (2 + 4)$ B $2 + 4 \div 8$
C $(2 + 4) \div 8$ D $2 \div 4 + 8$

**2** Write an algebraic expression for: *20 less than the sum of n and five.*

A $n + 5 + 20$ B $(n + 5) - 20$
C $n - 20 + 5$ D $20 - (n + 5)$

**3** Name the property shown: $1 \cdot 5 = 5$

A Commutative Property B Additive Identity
C Multiplicative Identity D Additive Inverse

**4** Simplify $(-3y^2 - 6y - 3)7$.

A $-21y^2 - 42y - 21$ B $-21y^2 - 6y - 3$
C $-21y^2 - 42y - 3$ D $-21y^3 - 42y - 21$

**5** Write an equation for the sentence: *three times the quantity x plus four equals sixteen.*

A $3x = 4 + 16$ B $3(x + 4) = 16$
C $3x + 4 = 16$ D $x + 3 \cdot 4 = 16$

**6** 26 students, or $\frac{2}{3}$ of the class, went on a trip. How many students are in the class?

A 38 B 40
C 39 D 41

**7** A class raised $\frac{1}{3}$ of the cost of a trip with a bake sale, then AED 300 more from a raffle. In total they raised AED 431. What is the cost of the trip?

A 394 B 392
C 395 D 393

**8** Does $12g - 12 = 4(3g - 3)$ have one solution, no solution, or is it an identity?

A one solution B exactly two solutions
C no solution D identity (all real numbers)

**9** Solve $5a - 3 = -8a - 94$.

A $-7$ B $-5$
C $-6$ D $-8$

**10** Solve $3|x - 2| - 9 = 18$.

A $\{-7, 11\}$ B $\{11\}$
C $\varnothing$ D $\{-11, 7\}$

**11** At a constant rate of 30 free throws in 6 minutes, how many throws in 24 minutes?

A 122 B 121
C 119 D 120

**12** Convert 90 km/h to metres per second. (1 km = 1000 m, 1 h = 3600 s)

A 24 B 25
C 27 D 26

**13** In the table for $y = -4x + 1$, what is y when $x = -5$?

A 22 B 20
C 21 D 23

**14** Is the function linear?

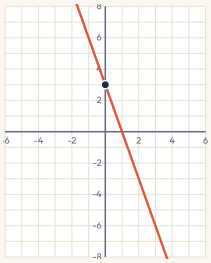
| x | 1 | 5 | 9 | 13 |
|---|---|---|---|---|
| y | -2 | -3 | -6 | -8 |

A Yes, the rate of change is constant B No, because y is negative
C No, the rate of change is not constant D Yes, because x increases by the same amount

**15** A club had 170 members in 2020 and gains about 2 members each year. Which equation gives the members y after x years?

A $y = 2x + 170$ B $y = 2x - 170$
C $y = 170x + 2$ D $y = 172x$

**16** Which equation matches the graph?



A $y = 3x + 3$ B $y = -3x - 3$
C $y = 3x - 3$ D $y = -3x + 3$

**17** Describe $g(x) = (x - 3) - 9$ compared with $f(x) = x$.

A translated 9 units left and 3 units up B translated 3 units right and 9 units down
C translated 3 units left and 9 units down D translated 3 units right and 9 units up

**18** Write an equation for the nth term of 4, 5, 6, ...

A $a_n = n + 3$ B $a_n = n + 4$
C $a_n = 4n + 1$ D $a_n = n - 3$

**19** 1 lifeguard is needed for every 40 swimmers (up to and including 40). How many lifeguards for 151 swimmers?

A 6 B 3
C 5 D 4

**20** For $g(x) = -2|x + 3| - 5$, find $g(4)$.

A $-20$ B $-17$
C $-19$ D $-18$

# Part B

**Questions 21–25 · Show your work · 8 marks each**

**21** Gym A charges a AED 120 joining fee plus AED 80 per month. Gym B charges a AED 200 joining fee plus AED 60 per month.
(a) Write an equation to find when the two gyms cost the same. (b) Solve it. (c) Which gym is cheaper for 6 months? Show why. **[8 marks]**

**22** The ideal temperature in a museum is 22 °C, give or take 3 °C.
(a) Write an absolute value equation for the minimum and maximum temperatures. (b) Solve it. **[8 marks]**

**23** On a map, 2 cm represents 15 km. Two cities are 9 cm apart on the map.
(a) Write a proportion. (b) Find the real distance. **[8 marks]**

**24** A taxi ride of 2 km costs AED 7 and a ride of 6 km costs AED 19. The cost is a linear function of the distance.
(a) Find the rate of change and explain its meaning. (b) Write the cost **y** in slope-intercept form. (c) What does the y-intercept mean? **[8 marks]**

# Part B

## Questions 21–25 · continued

**25** The rows of a theatre have 18, 22, 26, 30, … seats.
(a) Write an equation for the number of seats in row **n**. (b) How many seats are in row 15? (c) Which row has 70 seats? **[8 marks]**